

3.8 Polymerisation

Question Paper

Course	CIEA Level Chemistry
Section	3. Organic Chemistry
Topic	3.8 Polymerisation
Difficulty	Medium

Time allowed: 60

Score: /48

Percentage: /100

Question 1a

Addition polymers are made by the polymerisation of alkenes.

Explain what is meant by the term addition polymerisation.

[2 marks]

Question 1b

Trans-hept-2-ene can be made into an addition polymer.

i)

Give the displayed formula of trans-hept-2-ene.

[1]

ii)

Draw the structure of poly(hept-2-ene) showing two repeat units.

[1]

[2 marks]

Question 1c

Suggest why the polymer made from trans-hept-2-ene is not called poly(trans-hept-2-ene).

[2 marks]

Question 1d

Trans-hept-2-ene is a liquid at room temperature and pressure but poly(hept-2-ene) is a solid.

Explain why there is a difference in the states of trans-hept-2-ene and poly(hept-2-ene).

[3 marks]

Question 2a

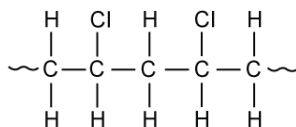
Polyvinyl chloride (PVC) is a polymer which is rigid enough for use as a drainpipe and flexible enough for plastic aprons.

State the IUPAC name of PVC.

[1 mark]

Question 2b

Fig 2.1 below shows a section of the PVC polymer.

Fig 2.1

i)

Draw the repeating unit of PVC.

[1]

ii)

Draw the structure and state the name of the monomer of the PVC polymer.

[2]

iii)

Write the equation to represent the formation of PVC from its monomer.

[1]

[4 marks]

Question 2c

The polymer poly(propene) is formed from the monomer propene.

i)

Draw the repeating unit of this polymer.

[1]

ii)

The properties of two different polymers, poly(methylpropene) and poly(ethene) are compared to each other. Complete Table 2.1 by placing an **X** in the correct box. You may place more than one **X** for each physical property.

[3]

Table 2.1

Physical property	Poly(methylpropene)	Poly(ethene)
Insoluble in water		
Unreactive		
High density		

iii)

Explain why the melting point of poly(methylpropene) is much higher than that of methylpropene.

[2]

[6 marks]

Question 2d

Polymers are unreactive compounds due to their long non-polar chain of saturated carbon-carbon and carbon-hydrogen bonds. Though this means that they are safe to use due it also means that they are non-biodegradable as they are not attacked by biological agents such as enzymes.

Table 2.2 shows three different ways in which polymers can be disposed of.

Table 2.2

Method of disposal	Disadvantage
Recycling	
Incineration	
Landfill	

State the disadvantage of each method of disposal to complete Table 2.2.

[3 marks]

Question 3a

This question is about different polymers and their uses.

Poly(tetrafluoroethene), also known as PTFE, is an addition polymer used as a non-stick coating in pans as it is very unreactive.

Draw the displayed formula of the monomer used to form PTFE.

[1 mark]

Question 3b

Draw two repeating units of poly(tetrafluorethene).

[1 mark]

Question 3c

The table shows the melting points of two polymers.

Polymer	Melting point / °C
Poly(ethene)	137
Poly(chloroethene) (PVC)	212

Explain the differences in melting point of these two polymers in terms of the intermolecular forces between the chains.

[5 marks]

Question 3d

Poly(ethene) bags pollute the environment for a long time because they are non-biodegradable.

Describe **two** pollution problems that are caused by non-biodegradable polymers.

[2 marks]

Question 4a

The structures of two different polymers, poly(ethene) and poly(propene) are shown in Fig. 4.1.

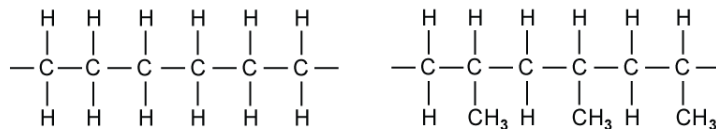


Fig. 4.1

Suggest **two** reasons why these polymers are unreactive.

[2 marks]

Question 4b

State **one** advantage and **one** disadvantage of the unreactivity of addition polymers.

[2 marks]

Question 4c

Another example of a polymer is polystyrene which is often used as packing material and electrical insulation.

Fig. 4.2 shows three repeating units of the polymer.

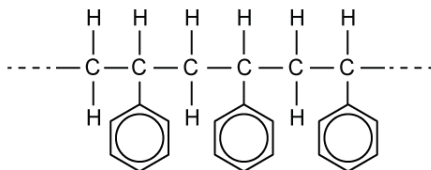


Fig. 4.2

Draw the structure of monomer **X** which is used to form the polymer shown in Fig. 4.2.

[1 mark]

Question 4d

Two different monomers can also join together to form an addition polymer.

Monomer **X** can react with monomer **Y** to form a polymer. Two repeating units of this polymer are shown in Fig 4.3.

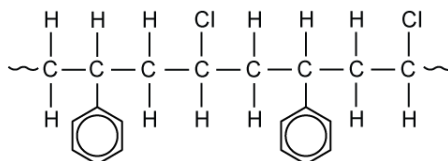


Fig. 4.3

- i)
Draw the structures of monomer **Y** which reacts with monomer **X** to form the polymer shown in Fig. 4.3.

[1]

- ii)
State the IUPAC name of monomer **Y**.

[1]

[2 marks]

Question 5a

Poly(but-1-ene) is insoluble in water so is used in applications such as making water pipes

Write an equation to show how poly(but-1-ene) is formed from the monomer but-1-ene.

[2 marks]

Question 5b

Not all polymers are insoluble in water. One example is poly(vinylalcohol) known as PVA. PVA is used in applications such as in the casing of detergent-filled capsules.

A section of the polymer chain of PVA is shown in Fig. 5.1.

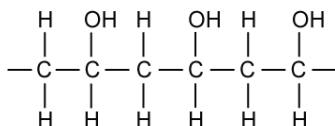


Fig. 5.1

i)

Draw the structure and give the IUPAC name of the monomer of PVA.

[2]

ii)

Suggest why PVA is soluble, but poly(but-1-ene) is not.

[2]

[4 marks]

Question 5c

Combustion can be used to dispose of poly(alkenes), such as poly(but-1-ene).

Give **one** advantage and **one** disadvantage to the environment of disposing of polymers by combustion.

[2 marks]

Question 5d

Poly(chloroethene), PVC, can also be disposed of via combustion but it produces toxic waste gases including hydrogen chloride.

Suggest a reagent that can be used to remove this toxic gas from the waste gases.

[1 mark]

Model Answers: Medium

1a

a) The term addition polymerisation means:

- The reaction where many monomers containing one (or more) C=C double bond; [1 mark]
- Join to form a long chain as the only product; [1 mark]

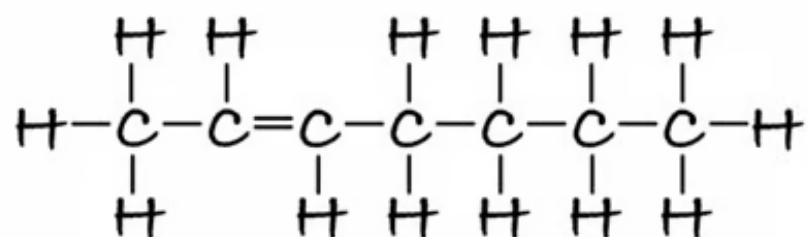
[Total: 2 marks]

- You must state that many monomers join to gain the first mark
- There is quite a lot of information that you need to include to get these marks, so this is worth learning
- It is important to note that only one product is formed

1b

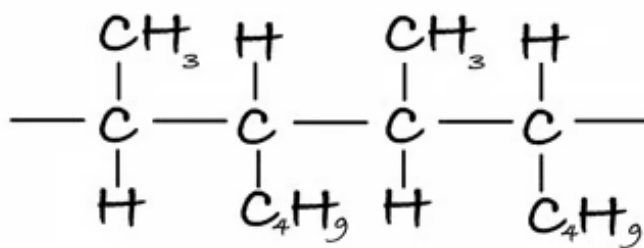
b)

i) The displayed formula of trans-hept-2-ene is:



- Correct structure with all bonds shown; [1 mark]

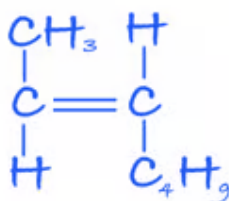
ii) The structure of poly(hept-2-ene) showing two repeat units is:



- Correct structure showing two repeat units **AND** continuation bonds; [1 mark]

[Total: 2 marks]

- When drawing the displayed formula of trans-hept-2-ene ensure you show all of the bonds
- The trans-isomer will have the hydrogen atoms attached to each carbon atom within the C=C double bond on the opposite side of the C=C double bond
- When drawing a polymer, you are best to draw the monomer showing the C=C bond like this:



- Then it is easier to construct the repeating unit by changing the C=C double bond to a C-C single bond and adding continuation bonds at either end
- You need to join two of these together to show two repeating units
- You would be awarded this mark if you drew both hydrogen atoms on the same side of the C=C double bond

1c

c) The polymer made from trans-hept-2-ene is not called poly(trans-hept-2-ene) because:

- The polymer does not contain a C=C double bond; [1 mark]
- (So) there are no geometric isomers; [1 mark]

[Total: 2 marks]

- In hept-2-ene, the groups attached to the C=C carbons remain fixed in their position
 - This is because free rotation of the bonds about the C=C bond is not possible due to the presence of a **π bond**
 - This leads to geometric isomerism
- When polymerised, the C=C double bond opens up to leave a C-C with σ -bonded carbons
 - The atoms / groups attached to these single, σ -bonded carbons are not fixed in their position due to the free rotation about the C-C σ -bond
 - Geometric isomerism does therefore not exist in the polymer
- Cis/trans nomenclature is used to distinguish between geometric isomers, so if no geometric isomers exist, there is no need for the prefix 'trans-' in the polymer name

1d

d) There is a difference in the states of trans-hept-2-ene and poly(hept-2-ene) because:

- Poly(hept-2-ene) contain more electrons; [1 mark]
- So it has greater / stronger instantaneous dipole - induced dipole / id-id / London forces; [1 mark]
- Which require more energy to overcome (causing it to have a higher boiling point); [1 mark]

[Total: 3 marks]

- When explaining differences in state, you need to explain why there is a difference in melting / boiling point
- For covalently bonded substances (except for giant covalent lattices) you need to refer to the intermolecular forces
- If you referred to the intermolecular forces as van der Waals forces, you would gain the mark but it is better to be specific as sometimes incorrect use of the term is penalised
 - Van der Waals forces refer to both permanent and id-id interactions

2a

a) The IUPAC name of PVC is:

- Poly(chloroethene); [1 mark]

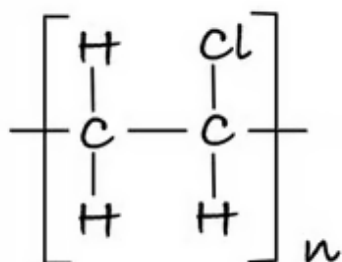
[Total: 1 mark]

- You must know that PVC is the common name for the polymer poly(chloroethene)

2b

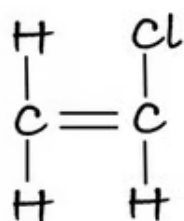
b)

i) The repeating unit is:



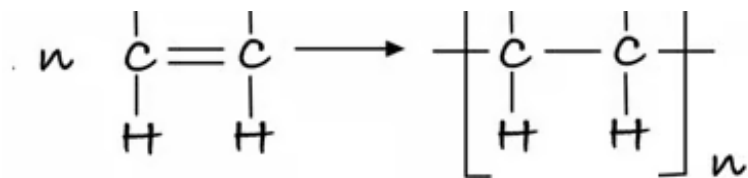
- Correct repeating unit drawn; [1 mark]

ii) The monomer is:



- Correct monomer drawn; [1 mark]
- The name of the monomer is chloroethene; [1 mark]

iii) The correct equation is:



- Correct equation drawn; [1 mark]

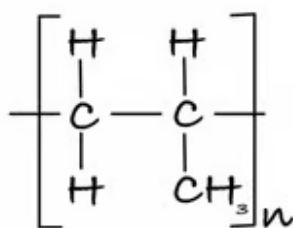
[Total: 4 marks]

- To deduce the repeating unit of a polymer, look for patterns that are repeated in the polymer section
 - If the polymer is made up of one type of monomer, then the repeating unit consists of two carbon atoms
- If the polymer is made up of more than one monomers, then the repeating unit may consist of more than two carbon atoms (4 if two monomers, 6 if three monomers etc.)
- **Note** that you **must** use square brackets and a subscript 'n' when drawing the repeating unit and also in the equations in front of the monomer

2c

c)

i) The repeat unit is:



- Correct repeating unit correctly drawn; [1 mark]

ii) The correctly filled table is:

Physical property	Poly(methylpropene)	Poly(ethene)
Insoluble in water	X	X

Unreactive	X	X
High density		X

- Each correct row; [1 mark]

iii) The melting point of poly(methylpropene) is much higher than that of methylpropene because:

- There are many more London dispersion / instantaneous dipole-induced dipole / id-id / van der Waals / intermolecular forces between molecules in poly(methylpropene); [1 mark]
- Which requires more energy to break down so has a higher melting and boiling point; [1 mark]

[Total: 6 marks]

- The type of forces acting between molecules of polymers are instantaneous dipole-induced dipole forces which are very strong since the hydrocarbon chains of polymers are very long and have many electrons
- This is why most polymers are **insoluble** (due to long non-polar hydrocarbon chain), **unreactive** and have higher melting and boiling points than other shorter molecules
- Since poly(methylpropene) is **branched** the polymers do not pack together as well as poly(ethene) causing there to be larger spaces in the structure
- Poly(methylpropene) is therefore **less dense** than poly(ethene) which is unbranched

2d

d) The disadvantages are:

- Recycling: Expensive / Difficult to separate the different polymers; [1 mark]
- Incineration: Production of pollutants / CO₂ / greenhouse gases

OR

A lot of energy / high temperatures are needed to combust the polymers; [1 mark]

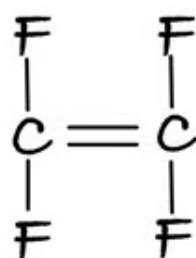
- Landfill sites: Last for many years/long time as they are not broken down by microorganisms; [1 mark]

[Total: 3 marks]

- You need to be able to evaluate the disposal of polymers
- Incineration is the combustion of polymers at very high temperatures
 - This produces CO_2 which is a greenhouse gas and contributes to global warming
 - Other toxic gases are produced as well

3a

a) The displayed formula of the monomer used to form PTFE is:



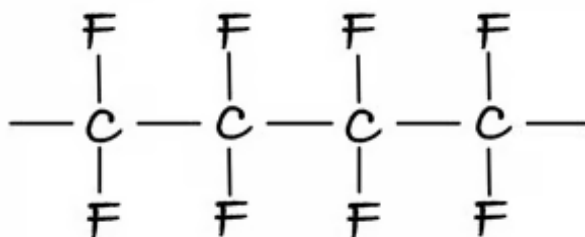
- Correct structure; [1 mark]

[Total: 1 mark]

- The monomer is given in the brackets of the name of the polymer, which is tetrafluoroethene
- The 'ethene' part of the name tells you that it has 2 carbon atoms, joined together by a double bond
- 'Tetrafluoro' tells you that there are 4 fluorine atoms attached to the carbon atoms - there is no need for numbers in this name as the carbon atoms will only form 4 bonds to other atoms, so there are no other alternative positions

3b

b) Two repeating units of poly(tetrafluoroethene) are:



- Correct structure showing two repeat units **AND** continuation bonds; [1 mark]

[Total: 1 mark]

- When drawing the displayed formula of PTFE ensure you show all of the bonds
- From the monomer as your starting point, change the C=C double bond to a C-C single bond and add continuation bonds at either end
- You need to join two of these together to show two repeating units

3c

c) Explaining the differences in melting point of these two polymers in terms of the intermolecular forces between the chains:

- Both polymers have instantaneous dipole - induced dipole / id-id / London dispersive forces; [1 mark]
- Poly(chloroethene) / PVC has stronger instantaneous dipole - induced dipole / id-id / London dispersive forces; [1 mark]
- Poly(chloroethene) / PVC also has permanent dipole - permanent dipole / pd-pd forces; [1 mark]
- Permanent dipole - permanent dipole / pd-pd forces are stronger than instantaneous dipole - induced dipole / id-id / London dispersive forces; [1 mark]
- Poly(chloroethene) / PVC requires the most energy to separate the chains
OR
Poly(ethene) requires the least amount of energy (to separate the chains); [1 mark]

[Total: 5 marks]

- You need to think about the difference in the structure between these two polymers to explain the difference in melting point
 - PVC is very similar to poly(ethene) but one of the hydrogen atoms has been replaced with Cl
 - As Cl is more electronegative than carbon and the molecule is not symmetrical, then PVC has a permanent dipole
- You need to compare the two polymers to be awarded the marks, just stating that poly(ethene) has weak id-id forces is insufficient
- Be careful with the terminology that you use, and be specific about what type of intermolecular force exists between these polymer chains

- When drawing the displayed formula of PTFE ensure you show all of the bonds
- From the monomer as your starting point, change the C=C double bond to a C-C single bond and add continuation bonds at either end
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3c

c) Explaining the differences in melting point of these two polymers in terms of the intermolecular forces between the chains:

- Both polymers have instantaneous dipole - induced dipole / id-id / London dispersive forces; [1 mark]
- Poly(chloroethene) / PVC has stronger instantaneous dipole - induced dipole / id-id / London dispersive forces; [1 mark]
- Poly(chloroethene) / PVC also has permanent dipole - permanent dipole / pd-pd forces; [1 mark]
- Permanent dipole - permanent dipole / pd-pd forces are stronger than instantaneous dipole - induced dipole / id-id / London dispersive forces; [1 mark]
- Poly(chloroethene) / PVC requires the most energy to separate the chains
OR
Poly(ethene) requires the least amount of energy (to separate the chains); [1 mark]

[Total: 5 marks]

- You need to think about the difference in the structure between these two polymers to explain the difference in melting point
 - PVC is very similar to poly(ethene) but one of the hydrogen atoms has been replaced with Cl
 - As Cl is more electronegative than carbon and the molecule is not symmetrical, then PVC has a permanent dipole
- You need to compare the two polymers to be awarded the marks, just stating that poly(ethene) has weak id-id forces is insufficient
- Be careful with the terminology that you use, and be specific about what type of intermolecular force exists between these polymer chains
 - **Remember:** Van der Waals forces refer to both instantaneous dipole - induced dipole forces and permanent dipole forces
 - You would not be awarded the first mark for stating either van der Waals forces or intermolecular forces

- Whenever you need to explain the melting points of covalently bonded substances, you need to refer to the types and strength of intermolecular forces that exist
- The strength of instantaneous dipole – induced dipole forces increase with the number of electrons

3d

d) **Two** pollution problems that are caused by non-biodegradable polymers are:

Any **two** from:

- Kill animals / ingestion can be fatal to animals; [1 mark]
- Animals can be caught in plastics e.g. fishing line; [1 mark]
- Combustion releases toxins / poisonous gases; [1 mark]
- Land-fill uses natural resources; [1 mark]

[Total: 2 marks]

- You should be able to answer this question using your knowledge of the environmental issues surrounding plastic and its disposal
- This list of accepted answers is not an exhaustive list, other appropriate examples are allowed

4a

a) **Two** reasons why these polymers are unreactive are:

- They are saturated / contain only single bonds; [1 mark]
- They are non-polar; [1 mark]

[Total: 2 marks]

- The 'suggest' command word should indicate that you will need to apply your knowledge to answer this question
- When studying polymers, you will have learnt that poly(alkenes) are unreactive so they are resistant to chemical attack, which is why they are non-biodegradable
- You may not have explored the reason why they are unreactive but you can use your knowledge from alkanes/alkenes to answer this
- You know that alkenes are more reactive than alkanes due to the C=C bond

- Polymers no longer have the C=C bond so they must be less reactive than the corresponding monomer
- You know that alkenes undergo electrophilic addition with HBr due to the polar bond within the HBr molecule
 - A lack of polar bonds in the polymers suggests that their molecules will be relatively unreactive

4b

b) **One** advantage and **one** disadvantage of the unreactivity of addition polymers are:

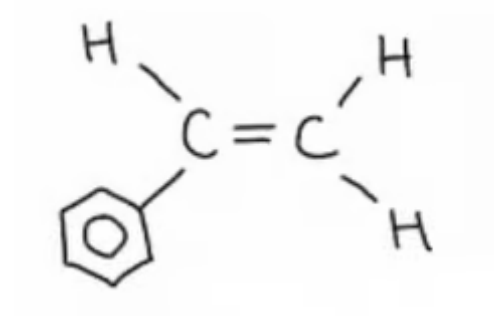
- Advantage: They are safe as they are chemically inert / durable; [1 mark]
- Disadvantage: They are non-biodegradable / produce pollutants when combusted / expensive / difficult to recycle them / difficult to decompose (in landfills / incineration); [1 mark]

[Total: 2 marks]

- Since polymers lack a double bond which is electron-rich, they will not be attacked by and attracted to electrophiles in the same way as alkenes are
- Like alkanes, polymers are therefore very inert and chemically unreactive
- This means that they will not undergo any dangerous reactions (for example with oxygen in the air) and are therefore safer to use but they can also be used as insulators
- However, the chemical unreactivity means that it is very difficult to decompose them
- They are non-biodegradable which means that microorganisms cannot break them down so they last for many years in landfill sites
- Polymers therefore need to be incinerated which involves combustion at very high temperatures
 - This produces CO₂ which is a greenhouse gas and contributes to global warming
 - Other toxic gases are produced as well
- Polymers can also be recycled however separating the different polymers from each other can be a difficult and expensive process

4c

c) The structure of monomer **X** are:



- Correct structure of monomer **X** drawn; [1 mark]

[Total: 1 mark]

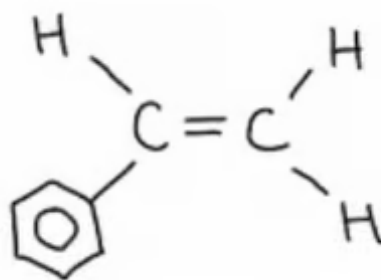
- Don't be put off this question by the unfamiliar ring attached included, this is a phenyl group but you do not need to know this to answer this question
- An addition polymer is formed when two monomers with a C=C bond join together
- An addition polymer has a backbone of carbon atoms and the monomer must contain at least two carbon atoms so that there can be a double bond between them
- In this question, we can see that two neighbouring carbon atoms each have at least one hydrogen atom
- One of the carbon atoms also has a phenyl group whereas the other one has a second hydrogen atom
- This means that the monomer consists of two carbon atoms which are bonded to each other by a double bond
- One of these carbon atoms is also bonded to a phenyl group

4d

d)

i) The structure of monomer **Y** is:

- The structure of monomer **Y** correctly drawn; [1 mark]



- Correct structure of monomer **X** drawn; [1 mark]

[Total: 1 mark]

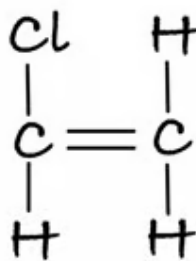
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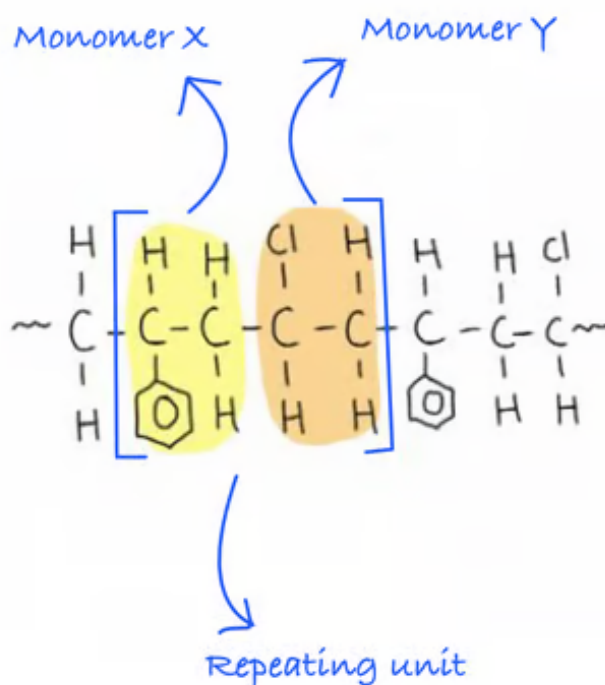


ii) The IUPAC name of monomer **Y** is:

- Chloroethene; [1 mark]

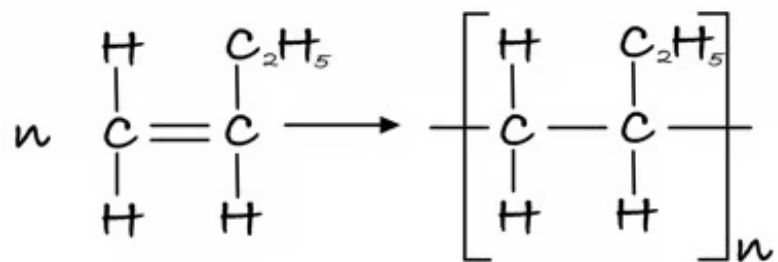
[Total: 2 marks]

- Firstly, identify the repeating monomer which consists of four carbon atoms
- Looking at the polymer in Fig. 4.3 we can also see that the phenyl group does not occur every other two carbon atoms anymore
- Furthermore, there is a Cl atom bonded to one of the carbon atoms
- This is because the polymer is made up of **two** monomers



5a

a) An equation to show how poly(but-1-ene) is formed from the monomer but-1-ene is:



- Correct structure of product **AND** continuation bonds either side; [1 mark]
- Rest of equation correct; [1 mark]

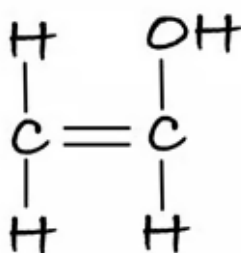
[Total: 2 marks]

- You must ensure that you use square brackets and 'n' to represent a very large number to be awarded the second mark
- When the polymer forms, the C=C double bond opens up and joins to another monomer which can be shown as a repeat unit

5b

b)

i) The structure and IUPAC name of the monomer of PVA are:

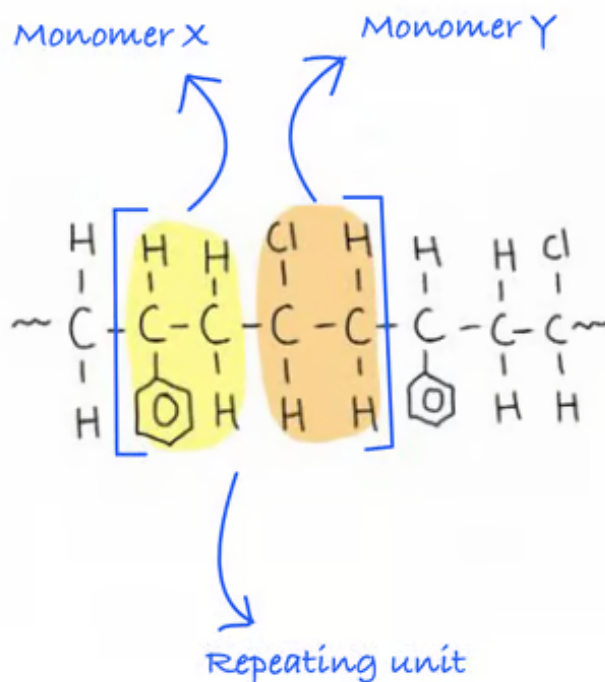


- Correct structure; [1 mark]
- Name: ethenol; [1 mark]

ii) PVA is soluble, but poly(but-1-ene) is not because:

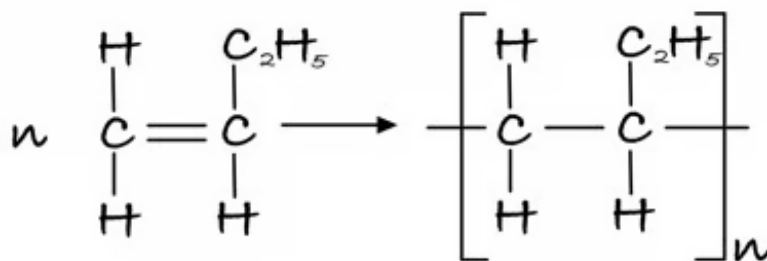
- PVA has OH / alcohol group(s); [1 mark]
- Which can form hydrogen bonds with water; [1 mark]

- Looking at the polymer in Fig. 4.3 we can also see that the phenyl group does not occur every other two carbon atoms anymore
- Furthermore, there is a Cl atom bonded to one of the carbon atoms
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5a

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- Correct structure of product **AND** continuation bonds either side; [1 mark]
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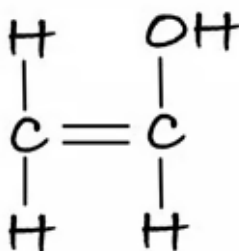
[Total: 2 marks]

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- When the polymer forms, the C=C double bond opens up and joins to another monomer which can be shown as a repeat unit

5b

b)

i) The structure and IUPAC name of the monomer of PVA are:



- Correct structure; [1 mark]
- Name: ethenol; [1 mark]

ii) PVA is soluble, but poly(but-1-ene) is not because:

- PVA has OH / alcohol group(s); [1 mark]
- Which can form hydrogen bonds with water; [1 mark]

[Total: 4 marks]

- To deduce the monomer, you can identify the repeat unit as the pattern of atoms attached to the carbon atoms repeats every 2 carbon atoms
- Draw one repeat unit and remove the bonds from either side of the carbon atoms, finally place a double bond between the two carbon atoms
- To name the monomer:
 - As the monomer has two carbon atoms, the prefix of its name is eth-
 - It has a double bond which is shown by -en- after eth-, so ethen-
 - It has an OH group, so it is an alcohol so will end in -ol
 - This gives the name ethenol
- Covalent structures will be insoluble in water unless they are polar molecules, or can form hydrogen bonds

- You should recognise that the -OH group has a very electronegative oxygen atom and an electron-deficient hydrogen atom, so will form hydrogen bonds with water

5c

c) **One** advantage and **one** disadvantage to the environment of disposing of polymers by combustion are:

- Advantage: Generates lots of heat which can be used to generate electricity
OR

They are not being disposed of in landfill; [1 mark]

- Disadvantage: Using up finite resources of oil to make more polymers
OR

Releases carbon dioxide which causes global warming

OR

Releases toxic gases; [1 mark]

[Total: 2 marks]

- Make sure that your answer refers to environmental issues, so an answer referring to cost would not be an acceptable answer
- Make sure you explain your answer, e.g. just stating that carbon dioxide is released is insufficient to gain the mark, you need to explain why this is an environmental problem

5d

d) A reagent that can be used to remove this toxic gas from the waste gases is:

- Any named base; [1 mark]

[Total: 1 mark]

- The key to this question is to recognise that hydrogen chloride is an acidic gas
- A neutralisation reaction is a suitable way to remove it, so any named base is acceptable, such as:
 - Sodium hydrogencarbonate
 - Potassium hydroxide